Architecture & Systems Engineering

Week 4: Modeling with DSMs and Modularization

Project Portfolio

Name

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Week 4 Project

Overview

In the fourth project activity of this course, you will continue to build on the system your team selected during Week 3. We will further study the system by decomposing it and reviewing its change propagation properties. The steps on the right will guide you through this process.

Note that some Scratch Pages are included at the end of this document for you to capture any ideas, sketches, etc. you have as you work through the project. These will not be assessed and you are not required to submit them with your project (but you may do so if you think they offer any additional insight into your thinking process!).

REQUIRED STEPS:

Step 1: Decompose the system.

Step 2: Analyze the system DSM.

Step 3: Identify the change propagation.

Step 4: Review and submit your project.

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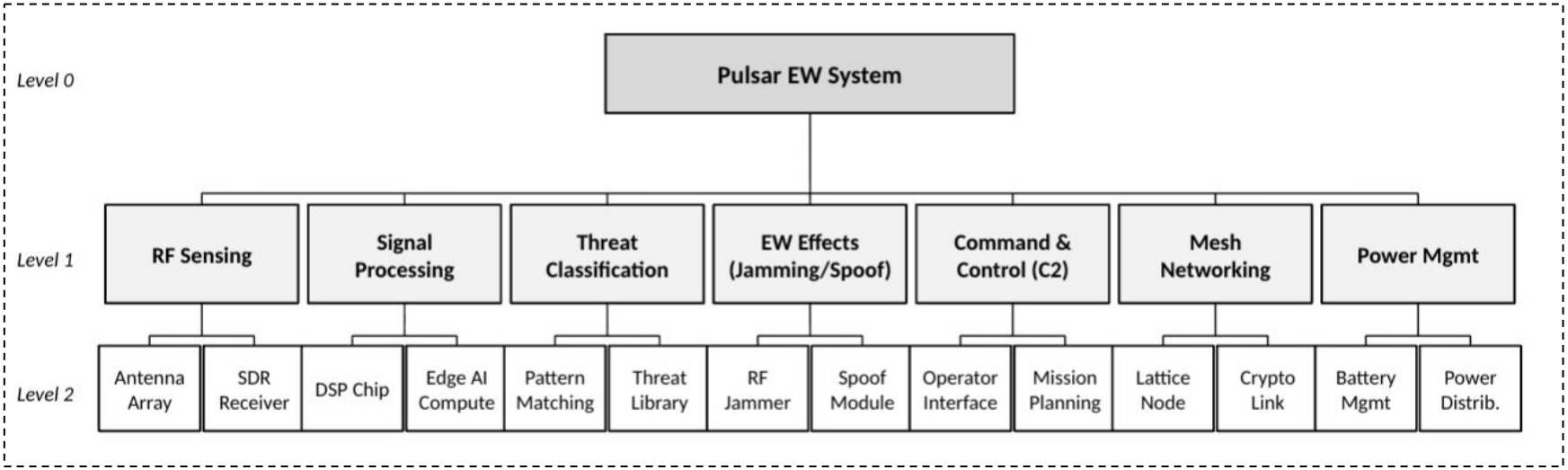
STEP 1: DECOMPOSE THE SYSTEM

For your system chosen in Week 3, develop a system decomposition of form. The system decomposition should include (1) a three level hierarchy of objects (including level zero) shown as a tree or an indented list, (2) 7±2 elements at level one, and (3) 7±2 elements for each set of children at level two. Please limit your analysis to one type of interaction (spatial, topological, energy flow, data, etc.).

Name of System: Pulsar - Electromagnetic warfare

System Decomposition:



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STEP 2A: SYSTEM DSM

Develop a N x N Design Structure Matrix of level two elements from Step 1 using the DSM Excel file provided in this project. Clearly label each row and column. Attempt to arrange the elements in the order of decomposition and mark the intended interaction using an X.

System DSM

Sort

Box

Dash

			A	B	C	D	E	F	G	H	I	J	K	L	M	N
Antenna Array		A	A	X												
SDR Receiver		B	X	B	X											
DSP Chip		C		X	C	X										X
Edge AI Compute		D			X	D	X					X				X
Pattern Matching	E	E				X	E	X								
Threat Library	F	F					X	F				X				
RF Jammer	G	G	X			X			G		X	X				X
Spoof Module	H	H	X			X				H	X	X				X
Operator Interface	I	I									I	X				
Mission Planning	J	J				X		X			X	J	X			
Lattice Node	K	K										X	K	X		X
Crypto Link	L	L											X	L		
Battery Mgmt	M	M													M	X
Power Distrib.	N	N													X	N
		O	A	B	C	D	E	F	G	H	I	J	K	L	M	N

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STEP 2B: SYSTEM DSM

Given the DSM developed in the previous step, attempt to create the modules of a new level one decomposition by clustering the elements of the system. How are the modules different from your level one decomposition in Steps 1 and 2a? What insights can you gain from your new decomposition?

System clustering:

Sort

Box

Dash

			A	B	C	D	E	F	J	I	G	H	K	L	M	N
Antenna Array		A	A	X												
SDR Receiver		B	X	B	X											
DSP Chip		C		X	C	X										X
Edge AI Compute		D			X	D	X		X							X
Pattern Matching	E	E				X	E	X								
Threat Library	F	F					X	F	X							
Mission Planning	J	J				X		X	J	X			X			
Operator Interface	I	I							X	I						
RF Jammer	G	G	X			X			X	X	G					X
Spoof Module	H	H	X			X			X	X		H				X
Lattice Node	K	K							X				K	X		X
Crypto Link	L	L											X	L		
Battery Mgmt	M	M													M	X
Power Distrib.	N	N													X	N
		O	A	B	C	D	E	F	J	I	G	H	K	L	M	N

Five clusters emerge after partitioning the DSM, consistent with the original Level 1 decomposition. Confirming the functional grouping was architecturally sound.

Edge AI Compute (D) and Mission Planning (J) are highly coupled bus elements connecting all clusters.

RF Jammer and Spoof Module (G, H) form a tightly coupled pair sharing identical inputs (Antenna, Edge AI, Mission, Operator, Power). They could be merged into a single Effects module.

Power Distrib. (N) is a cross-cluster interface element connecting Power to Sensing, Effects, and Networking.

5. **Crypto Link (L)** has only one connection (Lattice Node), suggesting it is a leaf module with low coupling.

Cluster color key:

- Sensing & Processing (A–D)
- Intelligence & Control (E,F,J,I)
- EW Effects (G,H)
- Networking (K,L)
- Power (M,N)

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STEP 3: CHANGE PROPAGATION

Given the DSM developed in Step 2A, identify a sorted list of level two objects (top five) which would create the largest change propagation chains in the system. For each, list the elements and up to three of the longest chains of propagation. When thinking about propagation, it is useful to remember that changes often exceed two generations but rarely five. Interconnected parts do not necessarily lead to endless change propagation, and propagation depends on many contextual factors in the real world.

Change propagating elements:

Element description	Number of Interfaces of the element	Change chain propagation [c1-c2-cX], [c1-c4-cX]
Edge AI	6	D → Pattern Matching → Threat Library → Mission Planning → RF Jammer/Spoof D → DSP Chip → SDR Receiver → Antenna Array D → Mission Planning → Operator Interface → RF Jammer/Spoof
Mission planning	6	J → Threat Library → Pattern Matching → Edge AI → Power Distrib. J → RF Jammer → Antenna Array → SDR Receiver → DSP Chip J → Lattice Node → Crypto Link
Power distribution	6	1. N → Edge AI Compute → Pattern Matching → Threat Library → Mission Planning 2. N → Edge AI Compute → DSP Chip → SDR Receiver → Antenna Array 3. N → RF Jammer/Spoof → Antenna Array
RF jammer	5	G → Power Distrib. → Edge AI → Pattern Matching → Threat Library G → Mission Planning → Threat Library → Pattern Matching → Edge AI G → Antenna Array → SDR Receiver → DSP Chip
Spoof module	5	H → Power Distrib. → Edge AI → Pattern Matching → Threat Library H → Mission Planning → Lattice Node → Crypto Link H → Antenna Array → SDR Receiver → DSP Chip

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